

"Physics as Information Processing" ~ Chris Fields ~

Lecture 3

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foreign it's July 13 2023 we're here in physics as information processing with the third lecture by Chris Fields so please check out the course website Coda document where you can submit questions check out the recordings and descriptions of all sessions and register if you want to participate in one of the discussion sections so thank you again Chris to you oh thank you and welcome to physics as information processing session three which is about Quantum reference frames so in this course we've talked about Quantum information Theory which we characterized as a general theory that describes the exchange of information in exchange of finite information um between two finite agents that are separated by some boundary and we've emphasize that this is a topological Theory not a geometric Theory so it makes no assumptions about an embedding space-time and last time I'll we showed how to characterize this boundary of script B as an array of in some finite and Quantum bits or qubits and how to characterize the interaction of the agents A and B with the boundary as an alternating cycle of of preparing and measuring these this the qubits in this array and we showed that their interaction could be represented as a sum of operators Each of which act on just one of the qubits and we also talked about the need for each of the agents A and B to choose a reference frame for their interaction with the qubits that defined uh which way was up so what the orientations of the qubits meant and emphasize that the two Observers uh A and B could choose different reference frames so different ways of measuring and preparing their qubits that use different meanings for the idea of up uh for each qubit and we ended the last session with a pointer forward toward the quantum formulation of the free energy principle which can be stated as the claim that interacting systems behave in a way that aligns their reference frames and I'll point it out at that time that what this means is that the two agents A and B approach entanglement asymptotically and that therefore the fep is a classical limit of the principle of unitarity which is the core principle of quantum theory so we'll see in this session why this is true so what we'll talk about today is what reference frames are in general how

they make observations and actions meaningful so how they give physics a semantics and then we'll talk about the free energy principle so what is a reference frame the idea that physics involves reference frames goes back at least a Galileo but was probably familiar well before that just not formalized in the way that Galileo formalized it so the key idea can be thought of uh in terms of motion and you're all familiar for example with riding a bicycle or driving a car and seeing the world as if it's passing by you at some velocity with some velocity and of course with respect to the world you're passing by it with some velocity and this idea that velocity is not absolute but that's relative to some Observer who measures it in some reference frame is the foundation for theories of Relativity so Galileo developed his theory of relativity which basically was just a way of converting from classical velocities measured by one Observer to classical velocities measured by another Observer uh that was moving with respect to the first one and this provided the foundation for Einstein's theories of relativity uh which modified Galileo's by adding the concept that The Observers have to change exchange information uh and can only exchange it at the speed of light which is finite so this idea in physics has a long history of course the core idea is that measurements are always with respect to something or other and there's something rather is a reference frame and different agents can choose different reference frames and this idea is very old here we have a picture of Anubis uh choosing a feather as the reference frame with which to rep to to measure the heart of the Pharaoh so this goes back to well before uh the Common Era this is about 2000 BCE so this is a very old idea that's been elaborated over the centuries as science has developed and in the 90s a fairly big change in how this idea was thought about occurred In classical physics reference frames were always thought of as mathematical abstractions and an observer and a reference frame were more or less identified by Galileo and that identification is carried forward into Einstein who thinks about observers for example sitting on photons and observers in spaceships and observers riding on railroad trains and this sort of thing and always that just means think of a reference frame that's doing something or other I'll a reference frame in classical physics is essentially just a coordinate system and these abstract coordinate systems can be employed in thought experiments and so they are a way to illustrate classical intuitions about motion and in the 90s as Quantum information theory developed it was realized that a reference frame actually consists of a physical object I mean even in the picture from the Egyptian tomb the reference frame is a physical feather and that means that the reference frame itself is described by quantum theory it's some Quantum system and if you think about reference frames that are familiar for example meter sticks for measuring distance or clocks

for measuring time or gyroscopes or batteries or even the Earth's gravitational fields which typically defines up for us those are all physical objects that can be described by quantum theory well this was an important insight of because it led to the realization that reference frames as Quantum systems are unique and uniqueness in quantum theory is very strong um there's the no cloning theorem which states that you can't reproduce exactly reproduce any unknown Quantum state and that applies to reference frames uh because they encode what Bartlett at all in that paper called non-fungible which just means information that can't be completely represented in any finite bit strain so unless you know the state of a reference frame you can't clone it uh since these reference frames um encode more than any bit strings worth of information in their Quantum phases of their unique and their states are unknown and the consequence of this is that if Alice wants Bob to use her reference frame to make a measurement she can't just send him a description she has to actually send the reference right and this has an important uh consequence for physics uh because it makes all replicate measurements approximate because observers obviously aren't using exactly the same physical objects as their reference frames that there's a deeper implication of this which is that Alice's ultimate Rock Bottom reference frame is actually implemented by her body by her brain and using any external reference frame requires having this internal reference frame that makes the external reference frame meaningful and usable so for example this thing is a piece of a meter stick a third of a meter stick but for me to use it to make measurements I have to actually have some concept of distance and that concept of distance is implemented by my nervous system and that implementation is my reference frame for length and it's what gives meaning to any external measurement of length it's what makes my measurements of length actionable by me using my effectors or action capabilities whatever they are so my internal reference frames are what make information I get from the world actionable and in in bateson's sense of differences that make a difference actionability is the core of semantics something is Meaningful if I can do something about it so this idea that we all uh contain embodied reference frames gives physics as semantics by giving observers or semantics and this is new uh this is an idea that never really appeared in physics before the measurements automatically had to have some kind of semantics but reference frames uh give semantics to measurements so here's an example of an embodied this is the reference frame that bacteria used for chemotaxis I.E to move toward molecules in the environment that they like like sugars and to move away from molecules that they don't like like acids and it's been worked out in great detail over Decades of experimental work and this is a rough schematic of the

reference frame has three parts of there's the measurement part which is conveniently located in the front of the bacterium and it consists of a bunch of receptors that probe the environment and they can bind molecules which are called ligands out in the environment and when they bind them they send a signal into the back of the bacterium and it turns out that there's a little miniature reference frame attached directly to these ligands which is this R and B system which controls the like the ligands the not the leggings the receptor sensitivity then the back of the reference frame is the action component and it consists of a set of molecules that control a motor a molecular motor that derives the flagella on the bacteria and the flagellum allows it to swim forwards or turn in some other direction and depending on how it spins uh the the creature either goes in the direction of something it likes or it stops and turns around and the stuff in the middle is a little molecular feedback loop that represents the amount of ligand in the environment that the bacterium expects to see so this is the middle of the reference frame and it defines the default value for what the bacterium is observing in its environment and if you're thinking in terms of the free energy principle or Bayesian inference then you'll immediately see the concentration of this molecule is encoding a prior probability the bacterium's prior probability for what its environment is like and the zero point the default value of that encoding is the ligand concentration in which the bacterium does nothing so a qrf in general can be characterized as an expected value which is here the concentration of this molecule key y phosphorylated plus a bunch of computation which is all the rest of the things in this diagram so this is a very simple qrf that's embodied by a bacterium here's a more complicated qrf that's embodied by eukaryotic cells who are more complicated cells like our cells and this is the reference frame that makes decisions about whether cells are going to divide so on the top you see the cell's membrane and there are all sorts of receptors sticking out of it that detect hormones and neurotransmitters and growth factors and this that and the other thing many of which are produced by other cells and those receptors um communicate via a lot of chemistry with uh the nucleus which is contained in another membrane and so serves as in a sense the outside upon which the reference frame is acting and what this reference frame does is control the transcription of large numbers of molecules from DNA so it basically controls what proteins the cell is making and those proteins end up doing something like restructuring the cytoplasm and actually driving cell division and the stuff in the middle of this which I've labeled the integrative Center are three molecules called RAF Mech and irk and they're connected by positive by bi-directional feedback loops and they basically negotiate the default value of the environment at which the cell so they're the middle

of the reference frame that represents the cell's expectation values about what's going on in its environment so you can imagine now what reference frames encoded by our nervous system look like they're extremely complicated and existed multiple scales from the scale of molecules in cells to the scale of large-scale connections between parts of our brains so how do we represent all this stuff uh there are many ways to do this but this is a particular particularly simple way of it assumes that whatever's in the middle which is here called C Prime the most General thing that the inputs can give information to and the most General thing that can give information to the outputs and most General here has a category theoretic definition um so it could be stated in a way that's equivalent across many different uh specific mathematical forms but if you're familiar with neural networks this will look to you a lot like the skeleton of a variational autoencoder and that's basically what it is and I don't want to deal with this formalism except to say that it exists and is very convenient uh except to say that we can attach this kind of formal diagram as a representation of a qrf to our representation of the boundary between two interacting agents and we can use these formal structures uh to drive the measurement operators that actually Implement physical changes on this boundary and I'll represent this whole business with just a triangle labeled Q for some qrf that's attached to the boundary and we can use this simple notation then to investigate what sorts of structures one gets when one starts to think about systems as implementing specific qrfs and here's the structure of the minimal system that we can really think of as an interesting Observer and this is a system that has some qrf that I've called e that looks at its environment the part of its environment that it can actually observe and act on and it defines a sector of the boundary how which mathematically is just the image of this qrfe that of the system can manipulate so this is the part of the environment or the part of the boundary that the system actually seeps and can do something about and for this for this Observer to be interesting it needs to be able to detect the difference between well it sees now and what it saw just then so that it can decide whether to take some action or other so it needs a memory and that requires another qrf that has some allocated piece of the boundary recall the boundary is where classical information gets written and this purely Quantum theoretic formalism so it needs a memory sector that that qrf can act on and then in order to do this computation and specifically in order to write to the boundary in order to actually flip bits on the boundary it needs free energy so the rest of its boundary gets devoted to extracting free energy from the environment and exhausting free energy waste heat in other words into the environment and so it has this nice little thermodynamic cycle that drives everything else and finally in order to tell now from just a second ago this the

information in memory it needs a clock that marks the information in memory as not current so as past information and if it can store multiple memory records it needs to be able to count and say this piece of remembered information is actually more recent than that piece over there so it has to implement a little internal clock and that clock is its internal time qrf and we've seen that qrfs act on boundaries this clock acts on a boundary but it doesn't act on B it acts on an internal boundary that's essentially the boundary between it the clock and the points that the ends of these qrfs that actually encode information that's either read from the environment or written to the environment so already this minimal Observer has a little bit of hierarchical structure which is that it's clock has to in some sense be higher up in the hierarchy than the parts the qrfs that interact directly with the environment so if you have a structure like this you face a metabolic trade-off for a fixed amount of energy that you can extract from the environment you can allocate it either into learning more about the environment or remembering more about what you learned before and so any system has to make this decision about how they're going to use energy are they going to devote it to learning more or remembering more and your laptop makes that decision memory is cheap for your laptop uh your brain makes that decision memory is much more expensive for your brain and all cells have to make that decision and the universal solution to this uh trade-off is course grading memory I.E remembering things in less detail than you saw them in the beginning uh And you have a comment uh yes uh I wanted to ask a question now before it gets too late so two or three slides ago um when we were looking at the diagram of the cone coon diagrams I was wondering if um the the classifiers identified with measurement and preparation lows correspond to sensory and action degrees of freedom in the earlier picture with bacteria and the the classifier in the Middle with that in that in that picture really correspond to the default chemical concentration that yes exactly okay perfect yeah that uh that crossing point in the middle is specifically where the default is encoded thank you for pointing that out okay okay uh so let's follow the consequences of this metabolic issue if you have a system that's a little bit smarter than our minimal system that can actually detect and measure the states of objects it has to do even more work because it has to encode even more qrfs so think of measuring the state of some object like a voltmeter in a laboratory if I tell you to go into the lab and tell me what the volt meter is reading then the first thing you have to do is find the voltmeter and your process of finding the voltmeter is independent of your process of reading what it's actually saying on its dial so in order to measure the state of an object you have to have two qrfs one that lets you find the object and then the other that lets you measure the state of interest of the

object and we call these the r sector for reference that lets you find the object in the P sector for pointer because old old instruments used to have pointers on dials that read analog numbers but what thinking about object identification tells you is that every object that you can identify uh leads to having to sector both the informative part of your boundary and the memory part of your boundary into two new sectors one to store information about object identification and one to store information about the states of those objects so this requires much more metabolism and again the solution is coarse green and typically what would be coarse green is our memory of the object that we've identified so what does this do to systems because making measurements costs energy and recording memories costs energy we have to make decisions about what qrf's to use at what time given our finite energy budgets we can't look at the world with all of our measurement capabilities at the same time because it's too expensive and now uh we see the effect of a theorem in which is that if two qrf's can't be deployed simultaneously for whatever reason uh that corresponds to them not commuting as operators and if they don't commute as operators they have to be implemented by compartments that are sufferable so compartments that only communicate classically the consequence of this theorem is that as soon as one has too many sensory and motor capabilities to run all at the same time given your energy budget uh you end up being compartmentalized into compartments that only communicate classically so you end up looking like this diagram where there are a bunch of components that do sensory motor Loops or memory one or the other and they can only communicate and they're controlled by some sort of attention system that selects which ones to activate depending on the amount of free energy that's available and clearly this clock function uh becomes part of this control system so the physics alone of any system that's acting as a fair that executes a fairly complicated interaction with its environment gives us this hierarchical compartmentalized structure without having to add any biological or cognitive science type assumptions at all so we could go in three directions from uh we can go deeper into the biology which we're going to do in October we can go deeper into the physics which we're going to do in August and September and we can go deeper into this issue of alignment which is what we're going to do today so there are four possibilities if you think about two agents that have qrf's that are interacting with across this boundary and the first one is Trivial if the the B agent the agent on the right side uh doesn't actually employ any qrf's if it just interacts with the boundary thermally uh then there is no alignment you're in a trivial situation where a can measure something about B but not in return if the B agent deploys a qrf that's much broader in scope than the a agent what we show

here in panel B then the a agent can learn something about the B agent but misses most of what the B agent is doing um in panel C we have agents with qrf's of the same size on the boundary but they're not lined up correctly so each one misses part of what the other is doing and finally in D we have the optimal situation where they have qrf's of the same size and they're actually lined up so each sees what the other is doing at least on this sector of the boundary and since the agents have to allocate some of the boundary to free energy Supply and can't look at that part then they can't tell what's happening uh across their entire boundaries they can only tell what's happening in this sector uh where they can actually deploy a qrf but let's now look at this interesting uh situation situation d uh where the qrf's are a large between a and b as I said that's the optimal channel for a and b and if we think about this we can write the variational free energy or the uncertainty or the prediction error $4A$ for the qrfx as the action that it sees I.E B's action the difference between that and what it expects β^2 so what it can predict about these actions and this is just of the notion of free energy uh from the free energy principle except translated into the situation where we have two qrf's and a can measure what's B has done but it can also use its qrf to predict oh because it's predicting the very state that it's going to observe next of the same sector that b is using now if we look at this diagram uh we can see easily that the way to minimize the variational free energy and therefore maximize predictive capability is for these two qrf's to actually be the same so if a is doing the same thing to the boundary that b is a can predict these next action perfectly because it's B is going to do whatever a is doing what the free energy principle tells us in this case is that a and b as they minimize their variational free Energies will approach having identical qrf's well again we have a theorem that tells us that if these qrf's are shared A and B are not separable but entangled and this is this is also fairly easy to see because it depends on this notion of non-fungibility uh if if x_a and x_b both encode non-fungible information which is quantum systems they do oh their states aren't clonable so they can't be copies of being sugar because one can't clone an unknown system so they can only be identical if they're actually identical systems oh this is exactly the same point made earlier about Alice having to actually send her qrf to Bob if she wants Bob to use her qrf minimizing vfe or Quantum systems is becoming entangled so the classical free energy principle which tells us to minimize vfa actually drives Quantum systems toward entanglement and that's why it's a special case of the principle of unitarity which just says that any isolated joint system is going to asymptotically approach entanglement so this raises an obvious question which is why all of us aren't entangled with our environments all the time uh and entangled with each other and the answer

is that entanglement is really hard for complex systems and this illustrates uh the the sense in which the fep is Moving Systems in two different directions at once the fep is Driving Systems to maintain their separability but it's also Driving Systems toward and as the system becomes more and more complex the difference between those two outcomes becomes larger and larger so recall from last time that the condition for separability is that the two systems have Dimensions that are much larger than the dimension of their boundary and this is the condition that allows them to have internal states to have them have states that aren't located on their boundaries and clearly to formulate the free energy principle uh you have to have internal States so if you're a complex qrf they're doing a lot of computation then you need a lot of internal degrees of freedom to implement their computation saving a lot of internal state so you have to have large dimensions so complexity favors separability as a system becomes more complex it has to have more internal States so it has to be in a sense more separable from whatever it's interacting with now think about this classic one if you have a complex system that's doing a lot of computation then it requires a lot of free energy and it has to dissipate a lot of entropy in the form of heat typically but something some form of energy transfer that its environment is going to see as noise as not informative so a complex system is working very hard to predict its environment's next state but by working very hard it's dumping a lot of entropy into its environment and putting in that extra entropy makes the environments Behavior even harder to predict so the more you work at predicting what the environment is going to do the more you yourself make the environment's behavior less predictable and this is the the the Paradox of the fep which is not really a paradox it's just what the physics does so we can think of the fep by Driving Systems to increase their distinctness from their environments or to self-evidence as Carl friston calls it um as uh driving this increased complexity that both makes predictions better and makes predictions Harder by transferring entropy into the so what this tells us is that the fep by will drive the development of systems with larger and larger complexity and this clearly has enormous implications for evolutionary theory which we'll talk about in October but I want to emphasize here that this is just the physics it's just what the fep does when it's forced to maintain a and before stopping here I want to consider one of aspect to this in particular which is language for example human language but any kind of language connecting any systems that are communicating with qrf's that are fairly close and language is a complex qrf that we basically share otherwise things like this interaction would be impossible but we also remain distinct systems so the question arises how do we manage to do this how do we manage to largely share a qrf all escaping becoming

entangled and this is a question that's largely been ignored in physics because physicists have tended to ignore language and just assume that observers can communicate classically so for example in talking about a Bell experiment which is a canonical experiment for measuring entanglement. The observers have to be able to share their results or they can't observe entanglement. Right? The observer, the observed result is a bunch of observational statistics that violate Bell's inequality and those statistics can only be computed given the results that both of the observers obtain so the observers have to share language to encode these results so how come the observers aren't entangled? Uh, how come there are still separable? What we end up appealing to is some sort of common cause in the past to explain why they share a language they both learned it from their culture so their culture is some sort of common cause but all such assumptions are fine-tuning assumptions so they all have basically the status of the value of something like the fine structure constant without which, uh, we wouldn't have atoms and so we wouldn't have us if this value has to be within very small limits of the value we actually observe and that's a case of fine-tuning. The universe seems fine-tuned in a way that allows us to exist and if it wasn't fine-tuned in that way we wouldn't exist and so we wouldn't observe anything but fine-tuning assumptions are a problem they're a pointer to something we don't understand and so I just want to leave you with the point that we have to be very careful when talking about interacting observers so in talking about two components of some system that jointly interact with some environment but also communicate classically and if you think back a few slides that's exactly the structure that we've shown that the physics forces any complicated system to have so next time what we're going to focus on is this very question of how we describe observers that are both interacting with the quantum system and communicating classically in some way so that session will be in August at the end of this month there's a discussion of this session with Ander and I encourage you all to use the interactive Q&A to ask questions you've been asking very good questions that I hope I've been managing to answer. Uh, so thank you very much and Daniel over to you. Oh, well we have a few minutes for some questions so maybe, um, and or if you want to bring up any questions first. Oh, I mean then go for it. I was curious if you could say any words on the slide where you had all the possible alignment of qubits and what you meant more by interacting thermally so when you have a trivial agent on one side what exactly does it mean to say that your interacting thermally if you think about a system that is only interacting thermally with its environment you end up thinking classically about for example an ideal gas that's in some sort of container and some of the molecules are very near the surface uh and their motions depend on the motions of all the

other molecules that are inside so there's a set of internal States that interact randomly with the external states that are near the boundary and so those external States interact randomly but notice in thinking in this way uh you've imposed a space-time embedded so you've actually used space as a way of separating the internal States from the boundary States and a system that uh if you drop this Assumption of this of a spatial boundary then a system that's only interacting thermally with its environment is going to look like a system that doesn't have any real internal States it's going to look like a system that just has boundary States and so a pure thermal interaction is going to start to look like entanglement so we're actually only really interested in systems that aren't interacting only purely thermally with their environments we're interested in systems that have enough internal structure that they're using that thermal energy to do something internally so to to give their internal Dynamics some particular shape so that their internal Dynamics is actually Circle circling around some well-defined attractor that requires some energy to maintain yeah I'm continuing with that question um I saw in the paper free energy principle for Quantum systems that if you're in that situation now when both α and β are non-trivial but perhaps β is larger than α 's local hidden variables so I wonder if there is a way to relate those by taking the limit of making α small and recover the thermal interaction presumably um if you let α become small you should recover that thermal interaction so to speak right yeah that's that's a very interesting comment actually so if you're a system and all of your environment's variables are effectively hidden to you oh then you can't do any information processing because you don't have any access to any of your system your environment's variables so yeah your interaction can only be thermal I think that's correct right thank you uh I guess yeah let's just so we since we're in in this in this topic and we don't have to context switch in that way um on the other hand you have noise right in in one of those other situations um I can't recall which one I guess I guess in the Reversed one right when when when α is larger than that of β so it encompasses more of the boundary decrease of Freedom those that are not overlapping a perceives us noise is that that's that's correct right well yeah it will be looking at at uh β 's thermal sector β 's free energy sector right so a may be interpreting in a sense what it's doing is informative and it and a may even be seeing what β is doing is informative oh but so so noise for β may be signal for α depending on on what how α 's qrf is structured oh but α is seeing more information than β is trying to send it so it's it's essentially the the reverse of the Hidden variable case there are variables in the other system that aren't telling you anything about what the other system is actually doing that that's very interesting in terms of um reading too much or too little between the lines

connecting too many dots or connecting not enough dots but these challenges where noise for one Quantum cognitive agent might be signal for another and vice versa those are the challenges that we want to address because to have a physics that doesn't recognize those situations is kind of taking the view from nowhere or not really addressing the complexity of a cognitive engagement yeah it's it's describing interactions between effectively systems that don't share the same language I have a question about the free energy principle here only in the last several years as the quantum free energy principle relationship come into light so maybe could you just share a little bit about what incremental or Quantum advances enable that to happen or what degrees of freedom were pre-existing or added to enable work arising from perception and action in neural systems to also so naturally enter this context with the quantum well I I think the at least from a historical point of view uh that connection was made based on the analogy between a Markov blanket and a holographic screen so these are structures developed in different parts of physics that essentially serve the same function of limiting the information that one system can have about another system so uh it's that functional analogy between a Markov blanket defined classically and a holographic screen defined in quantum theory that uh made it seem obvious that we could translate the free energy principle into a Quantum theoretic principle yes in live stream 40 we we talked a lot about this but I think that remains one of the most remarkable advances inside or outside the free energy principle literature based upon this functional acknowledgment of boundaries and the conditions that they entail which again are the challenges that we want to address lest we ignore boundaries right yeah precisely I mean the free energy principle is all about what happens when you have a boundary the waste Heat is this the actual thermal exhaust from our CPU or how can we think about the Heat being released topologically rather than geometrically like the location where it's released what does it mean to consider physical processes topologically that we might otherwise just use a geometric way of thinking about oh well you basically put your finger on it and that you don't you're you're not worried about the spatial difference between the geometric spatial difference between oh where one kind of information flows through the boundary and where another kind of information flows through the uh and you can think about this in bar called blanket terms uh in the Marco blanket is just a network and some of the the causal arrows flowing through that blanket Network are effectively transmitting noise and you can you can get in the same way with respect to a holographic screen if you're doing a certain amount of information processing and this this brings up the whole issue of oh how how efficient is a system in its information processing

how close is it to the the land hour limit basically can it extract more more uh energy from some bits than it has to use to modify the states of other bits oh that's what this this if you will metabolic question is all about uh in the case of an organism is it can I get more energy from some of the chemistry that I have access to then I need to use to modify the chemistry that I need to modify and if not then life stops right everything grinds to a halt now can the heat exhaust informationally be useful for another agent somehow I mean can't we recover heat in a steam engine and use it to do useful work not to the point of a perpetual motion machine but how can we recover exhaust and then what does the perpetual motion machine look like well the the environment is the system sitting over on the other side of the that is uh absorbing whatever waste heat is generated by the let's call it Alice that we're interested in so that environment of maybe and and Alice can't tell this it may be divided up into lots of different compartments some of which are basically sucking in Alice's waste heat and doing something with it so this is really a question about how is the environment structured and what uh operations does the environment execute with the input that it gets how does it interpret that input and what would uh let's get back to the this gets back to this question about uh how do we talk about multiple agents that are communicating with each other right instead of talking about uh and environment that we don't characterize at all how do we talk about an environment that includes some specified or preferred systems that are treated as or known to be other agents so so uh let me get back to your question about perpetual motion machine uh notice that In classical thermodynamics there's always the assumption that there's a big environment out there you know the rest of the universe or whatever the the whole environment that's not on Earth that is an arbitrarily large source of or it's not an arbitrarily large source of free energy but it's a large source of free energy and it's a large sink for entropy it kind of arbitrarily large sink for entropy so the second law of Thermodynamics says the entropy of this entire system counting uh the the whole surrounding environment doesn't decrease so it can increase and and still satisfy the second law so this gives you the the classical you know 19th century heat death of the universe idea and so you're you're a proposed machine sitting somewhere embedded in this gigantic environment that can provide some finite amount of free energy but can exhaust effectively an arbitrary amount of entropy so any any machine has to dissipate entropy as it runs to to fill up this kind of arbitrary sync of entropy that's the classical rest of the universe now compare that to the picture in quantum theory right the picture in quantum theory is the universe by definition is an isolated system so it's total information content doesn't change right it's evolving unitarily unitary operations

preserve information so they're just moving information so from any uh from the point of view of any local the local systems getting information from its environment so it sees its environment losing information so gaining entropy what this tells us is that that entropy is in fact not a globally definable quantity entropy is always defined relative to some Observer some division some boundary and this is this is another kind of big change in the last few decades and has led to for example in Tropic definitions of time now my local external time Arrow is whatever Direction I see entropy increasing the most so again I would refer to people like ravelli uh and Max tobark who developed many of these ideas awesome all right thank you Chris and under looking forward to seeing everyone's questions submitted and their preparation and participation and measurements in the discussion section so thank you until next time all right thanks again Daniel uh thank you under uh we'll see you later bye